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Slide material is largely sourced from Slack & Bandon-Jones (2019) *Operations Management.* Ninth Edition (Pearson), and Heizer, Render & Munson (2019) *Operations Management*. Twelfth Edition (Pearson)

Operations Management – Introduction

Lecture 1

Dr Roberto Chavez

SWINBURNE UNIVERSITY OF TECHNOLOGY

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

OUTLINE

- What is operations management(OM)?
- Why study OM?
- The process view
- Types of operations: The 4Vs model
- Current challenges

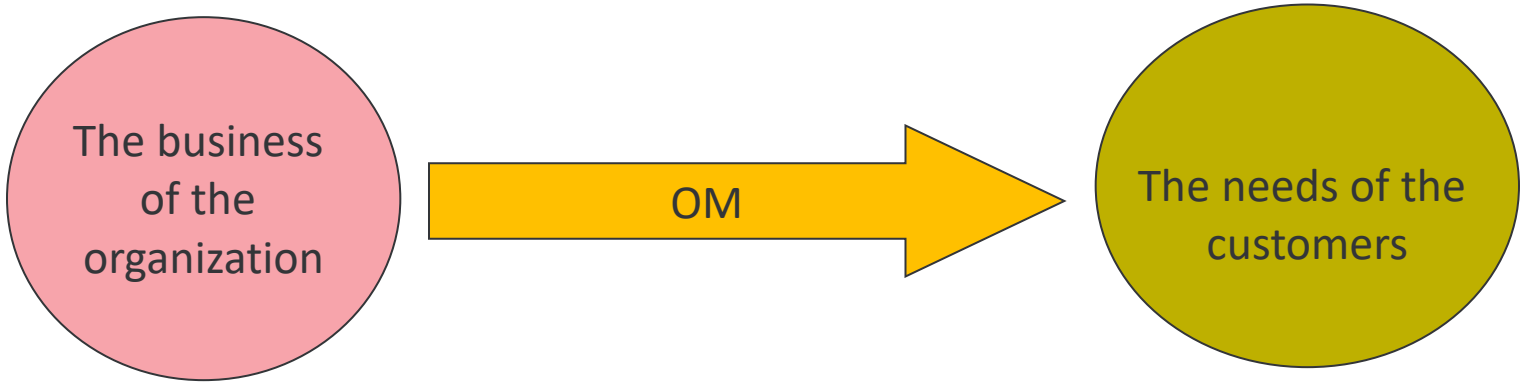
WHAT IS OM?

Operations management (OM) is the set of activities that create value in the form of goods and services by transforming inputs into outputs

Source: Heizer, Render & Munson (2017)

OM AND THE CUSTOMER

"Any customer, in any industry, in any market wants stuff that is both cheaper and better, and they want it yesterday"



Source: Ridderstrale and Nordstrom, 2000

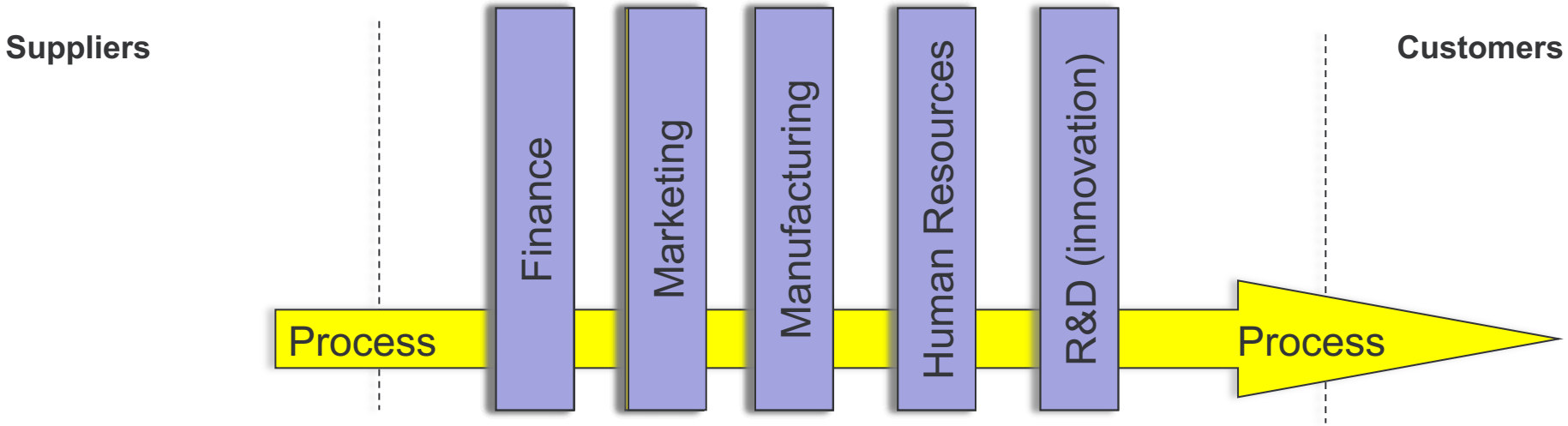
WHAT IS OM FOR AMAZON?

Source: https://www.reviewjournal.com/business/business-columns/real-estate-insider/amazon-warehouses-sell-for-big-money-in-southern-nevada-2587232/

THE PROCESS VIEW

What do you think about this statement:

"Successful products are the results of processes, not functions!"



Operations Management

Supply Chain Management

Sources: Slack et al. 2007

THE PROCESS VIEW: ENTREPRENEURSHIP OR OM?

IS OM THE SAME FOR EVERY BUSINESS? TYPES OF OPERATIONS: TYPOLOGY

- Business (operations) are similar in that they transform inputs into outputs; however, they can be differentiated through specific product/service characteristics.
- 4Vs model (typology)
  1. Volume
  2. Variety
  3. Variation in the demand
  4. Visibility of the operation with the customer

Sources: Slack et al. 2007

VOLUME

Processes

Low repeatability
Low specialization
Low systemization

Low Volume *High*

Processes

High repeatability
High specialization
High systemization

Sources: Slack et al. 2007

VARIETY

Processes

High repeatability
High specialization
High systemization

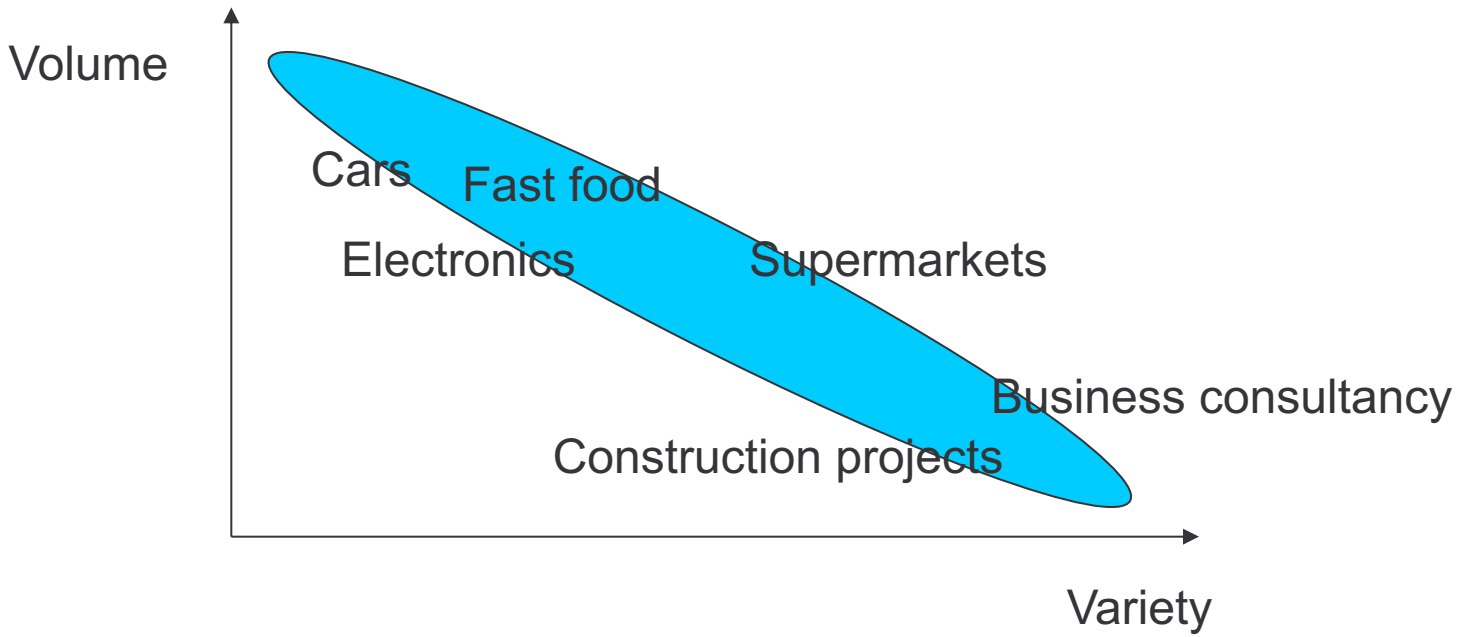
Low Variety *High*

Processes

Low repeatability
Low specialization
Low systemization

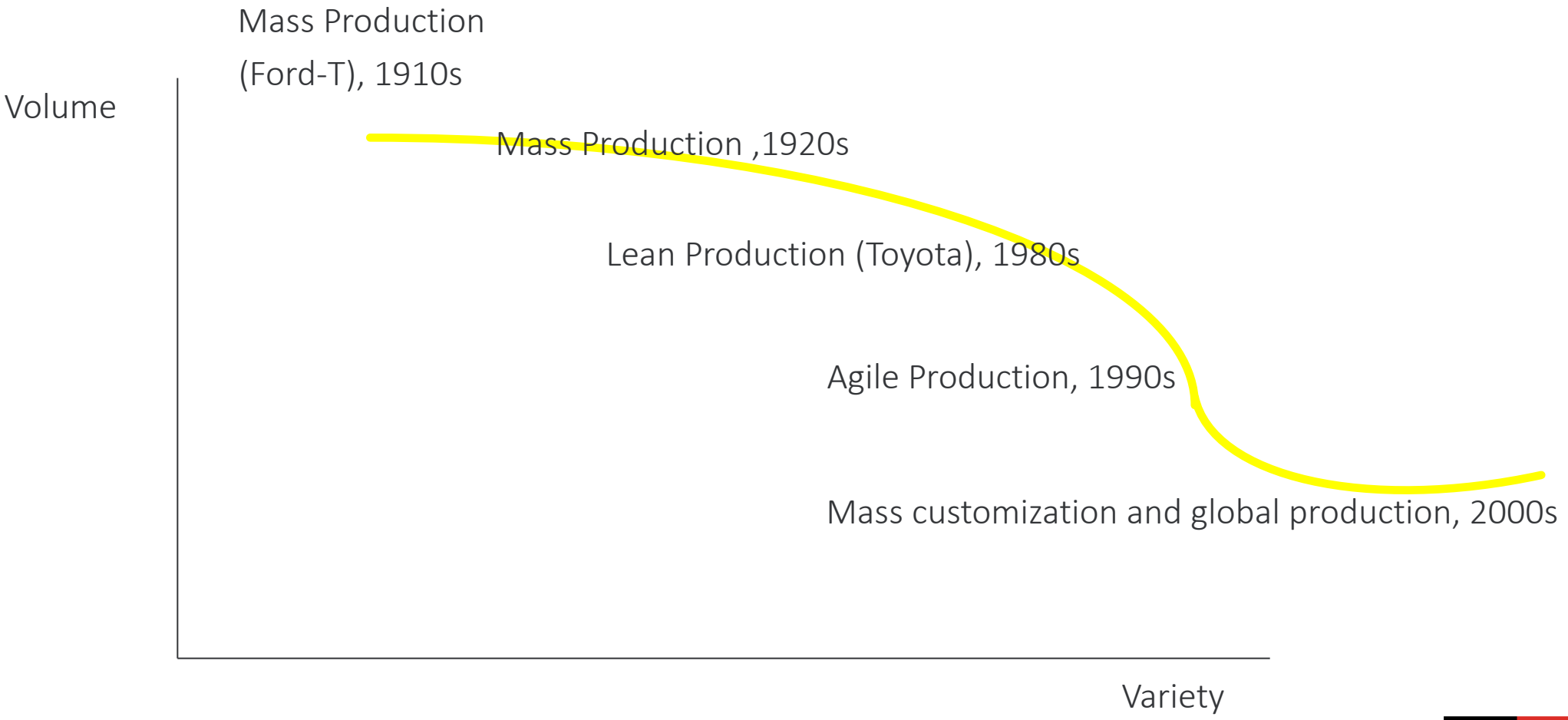
Sources: Slack et al. 2007

INDUSTRY TYPE:
THE VOLUME-VARIETY TRADE-OFF



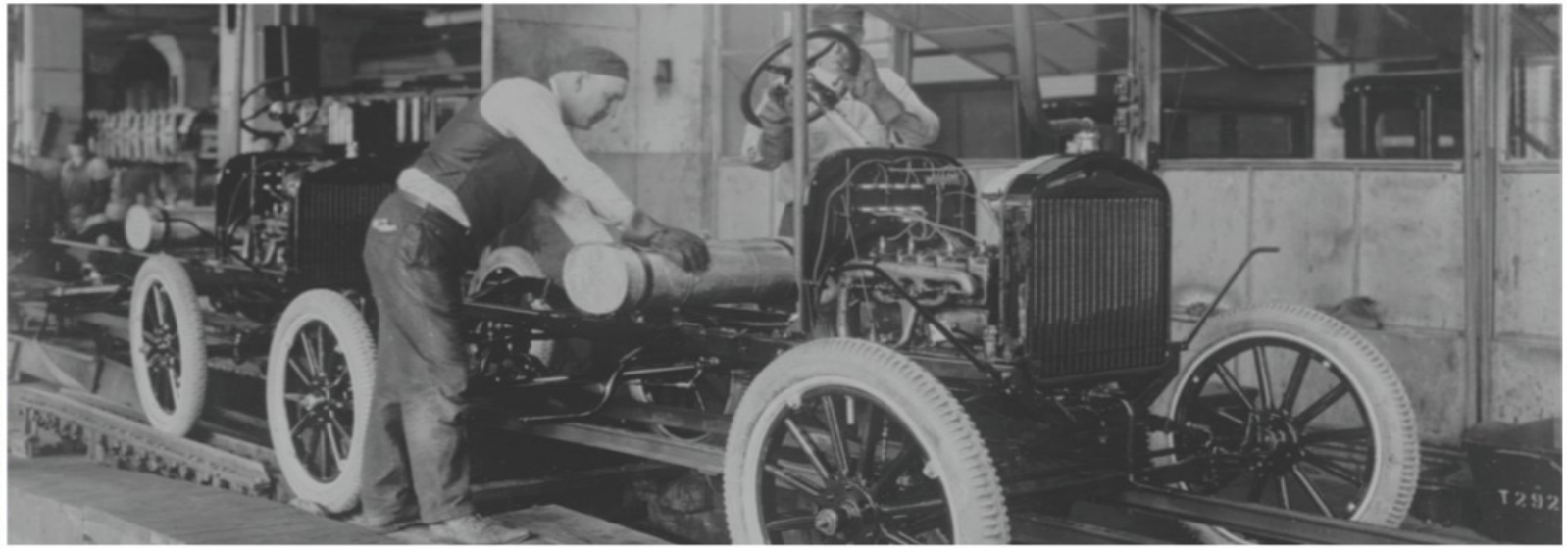
Sources: Slack et al. 2007

INDUSTRIAL EVOLUTION:
THE VOLUME-VARIETY TRADE-OFF



Sources: Slack et al. 2007

SIGNIFICANT EVENTS IN OM



Everett Collection/Newscom

Cost Focus

Early Concepts 1776–1880
Labor Specialization (Smith, Babbage)
Standardized Parts (Whitney)

Scientific Management Era 1880–1910
Gantt Charts (Gantt)
Motion & Time Studies (Gilbreth)
Process Analysis (Taylor)
Queuing Theory (Erlang)

Mass Production Era 1910–1980
Moving Assembly Line (Ford/Sorensen)
Statistical Sampling (Shewhart)
Economic Order Quantity (Harris)
Linear Programming
PERT/CPM (DuPont)
Material Requirements Planning (MRP)

Quality Focus

Lean Production Era 1980–1995
Just-in-Time (JIT)
Computer-Aided Design (CAD)
Electronic Data Interchange (EDI)
Total Quality Management (TQM)
Baldrige Award
Empowerment
Kanbans

Customization Focus

Mass Customization Era 1995–2005
Internet/E-Commerce
Enterprise Resource Planning
International Quality Standards (ISO)
Finite Scheduling
Supply Chain Management
Mass Customization
Build-to-Order
Radio Frequency Identification (RFID)

Globalization Focus

Globalization Era 2005–2020
Global Supply Chains
Growth of Transnational Organizations
Instant Communications
Sustainability
Ethics in a Global Workforce
Logistics

Source: Heizer, Render & Munson (2017)

VARIATION (DEMAND)

Processes

High repeatability
High specialization
High systemization

Low Variation *High*

Processes

Low repeatability
Low specialization
Low systemization

Sources: Slack et al. 2007

VISIBILITY (CUSTOMERS AND SERVICE)

Processes

High repeatability
High specialization
High systemization

Back office operation in a bank

Low Visibility *High*

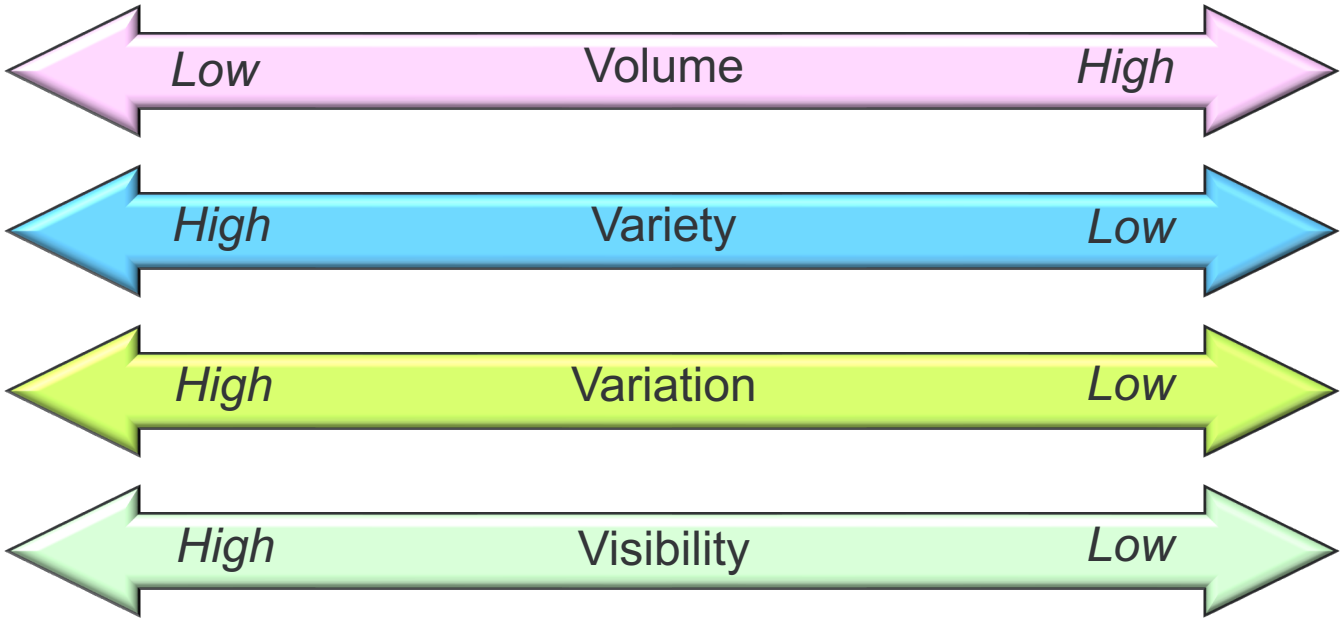
Processes

Low repeatability
Low specialization
Low systemization

OPERATIONS FIT: THE 4VS MODEL APPLIED

1) Think about a successful company you are familiar with and try to classify its operations considering the 4 Vs model

2) Is there a fit between the nature of the business (operations - 4Vs) and the type of operations deployed (high or low specialization, repetition and systemization)



OM: CURRENT CHALLENGES

- Globalization
- Rapid product development
- Mass customization
- Sustainability
- Natural disasters
- Poverty
- Education

1) Pick any of these topics and discuss how is it related to OM? Discuss in groups.

Source: Heizer, Render & Munson (2017)

DIGITAL TRANSFORMATION

- Digital transformation in the OM context

- Discuss in groups how digital transformation and OM can contribute to sustainability issues? Illustrate your answer with an example.

UNIT STRUCTURE

Strategic decisions
Operations management
Operations strategy

Process decisions
Process and layout design
Product design

Capacity decisions
Capacity planning
Inventory management

Operations decisions
Lean management
Quality management
Supply chain management
Sustainability management

Source: Krajewski et al. 2000

REFERENCES

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